Predicting Mental Health Risk Among University Students

Jawaher Alzahrani
Soorya Sathish
Sudhersan Srinivasan Kunnavakkam Vinchimoor
Osama Alshehri
Arizona State University

Abstract

Mental health issues among university students have grown due to academic pressure, lifestyle changes, and social challenges. This project aims to predict students' mental health risk levels (high-risk vs. low-risk) using machine learning models applied to survey-based data. The dataset includes 7,022 student records with 20 features covering demographic, academic, and lifestyle-related factors. A composite risk score was created by averaging stress, anxiety, and depression levels and then converted into a binary classification label. A major challenge in this task is class imbalance, which caused baseline models to achieve high accuracy but poor performance in detecting high-risk students. To address this issue, SMOTE was applied within the training pipeline to improve learning for the minority class. Multiple models were evaluated, including Logistic Regression, Random Forest, Support Vector Machine (SVM), Gradient Boosting, and XGBoost. The best-performing model was dynamically selected based on the highest Test F1-score, with Logistic Regression achieving the best performance. The model achieved an F1-score of 0.423 using the default threshold, which improved to approximately 0.457 after threshold tuning, along with enhanced recall in detecting high-risk students.

1 Introduction

1.1 Background

Mental health has become an increasingly important issue among university students. Many students face academic pressure, financial stress, and lifestyle changes, which can contribute to conditions such as anxiety and depression. These challenges can affect both academic performance and overall well-being, making mental health a critical area of concern in higher education.

1.2 Problem Definition

This project focuses on predicting mental health risk among students using machine learning. The goal is to classify students into two categories: high-risk and low-risk. To define the target variable, a composite score was created by averaging stress level, anxiety score, and depression score. This score was then converted into a binary label using a predefined threshold, allowing the problem to be treated as a classification task.

1.3 Importance and Motivation

Early identification of students at high risk is essential for providing timely support and intervention. Universities can benefit from predictive models that help identify at-risk students before their condition worsens. From a machine learning perspective, this problem is challenging due to class imbalance. In this context, misclassifying high-risk students (false negatives) is more critical than misclassifying low-risk students. Therefore, evaluation metrics such as recall and F1-score are more suitable than accuracy.

1.4 Existing Literature

Previous research has demonstrated the effectiveness of machine learning in predicting mental health conditions using behavioral and demographic data. For example, Chawla et al. (2002) introduced the SMOTE technique to address class imbalance by generating synthetic samples for the minority class, which improves model performance in imbalanced datasets.

In addition, recent studies have applied machine learning models to predict mental health conditions among university students using survey-based data, showing that factors such as stress, sleep quality, and academic performance are strong predictors. Furthermore, research has shown that machine learning algorithms can effectively identify depression-related patterns using health and demographic data.

However, a common challenge highlighted across these studies is class imbalance, which often leads to models achieving high accuracy but poor performance in detecting high-risk individuals. Therefore, handling class imbalance is essential for building reliable and effective mental health prediction systems.

1.5 System Overview

The proposed system follows a standard machine learning pipeline:
Data → Preprocessing → Modeling → Evaluation
The workflow includes the following steps:

- Data loading and inspection
- Target variable construction
- Data preprocessing (encoding and scaling)
- Model training using multiple classifiers
- Performance evaluation using appropriate metrics

1.6 Data Collection

The dataset used in this project was obtained from a publicly available student mental health survey dataset on Kaggle. It includes 7,022 student records with 20 features covering demographic, academic, and lifestyle-related factors.

1.7 Components of the Machine Learning System

The system consists of several key components:

- Feature processing: handling missing values, encoding categorical variables, and scaling numerical features
- Target construction: creating a composite risk score from stress, anxiety, and depression
- Model training: applying multiple classification models
- Imbalance handling: using SMOTE within the pipeline to address class imbalance
- Evaluation: measuring performance using accuracy, precision, recall, F1-score, and ROC-AUC

1.8 Brief Experimental Results

Initial baseline models achieved high accuracy but showed poor performance in identifying high-risk students, indicating the presence of a class imbalance problem. The models tended to favor the majority class, resulting in low effectiveness in detecting at-risk individuals.

To address this issue, SMOTE was applied within the training pipeline to balance the class distribution. This significantly improved the model's ability to learn patterns associated with high-risk students. After applying SMOTE, all models showed improved performance, particularly in terms of recall and F1-score. The best-performing model was dynamically selected based on the highest Test F1-score, with Logistic Regression achieving the best performance.

Further optimization, including threshold adjustment, enhanced the model's performance. The F1-score improved from 0.423 to approximately 0.457 after threshold tuning, along with increased recall for high-risk cases, making the model more effective in identifying at-risk students. These results demonstrate that handling class imbalance, combined with dynamic model selection and threshold tuning, is essential for building effective mental health risk prediction systems.

2 Definitions and Problem Statement

2.1 Overview

In this section, we explain the dataset we used, what exactly we are trying to predict, and how the problem is defined. The main goal of this project is to use student survey data to identify whether a student is at risk in terms of mental health.

2.2 Dataset Description

The dataset used in this project is based on survey responses collected from university students. Each row represents a student, and each column represents a different feature related to their academic, personal, or mental health condition.

The dataset mainly includes values related to stress, anxiety, and depression levels, along with some other student-related information. Since the data comes from survey responses, it may have some variation and noise, so it needs preprocessing before applying machine learning models.

2.3 Prediction Target

The main target variable in this project is mental health risk. This is not directly given in the dataset, so we define it ourselves. We calculate the average of stress, anxiety, and depression scores for each student. Based on this average value, we classify students into two categories:

- High-risk: students who may need attention or support
- Low-risk: students who are within a normal range

This converts the problem into a binary classification problem.

2.4 Problem Formulation

Given a dataset of student features X, the goal is to learn a function that can predict the corresponding label y, where y represents whether a student is high-risk or low-risk. In simple terms, we want to build a model that takes student data as input and predicts their mental health risk level.

2.5 Challenges and Constraints

There are a few challenges in this problem. One major issue is class imbalance, where the number of low-risk students is higher compared to high-risk students. This can make the model biased toward predicting the majority class.

Another challenge is that survey data can sometimes be inconsistent or noisy, which can affect model performance. Also, the model should work well on new unseen data, not just the training data. Because of these reasons, careful preprocessing and model selection are important for this task.

3 Overview of Proposed Approach

3.1 Overview of the System

In this section, we explain the approach used in this project to predict mental health risk among students. The process follows a basic machine learning pipeline.

First, we understand the dataset and perform exploratory data analysis (EDA) to see patterns in the data. Then we apply preprocessing steps like encoding and scaling. After that, we train different models and compare their performance. Finally, we evaluate the models using different metrics.

3.2 Exploratory Data Analysis (EDA)

Before building the models, we explored the dataset to understand how the data looks and how different features are distributed.

We checked basic patterns in the data and also created a few simple visualizations to make it easier to understand. From this, we noticed that the dataset is not perfectly balanced, and some categories have more values than others.

These observations helped us understand the dataset better before moving to modeling.

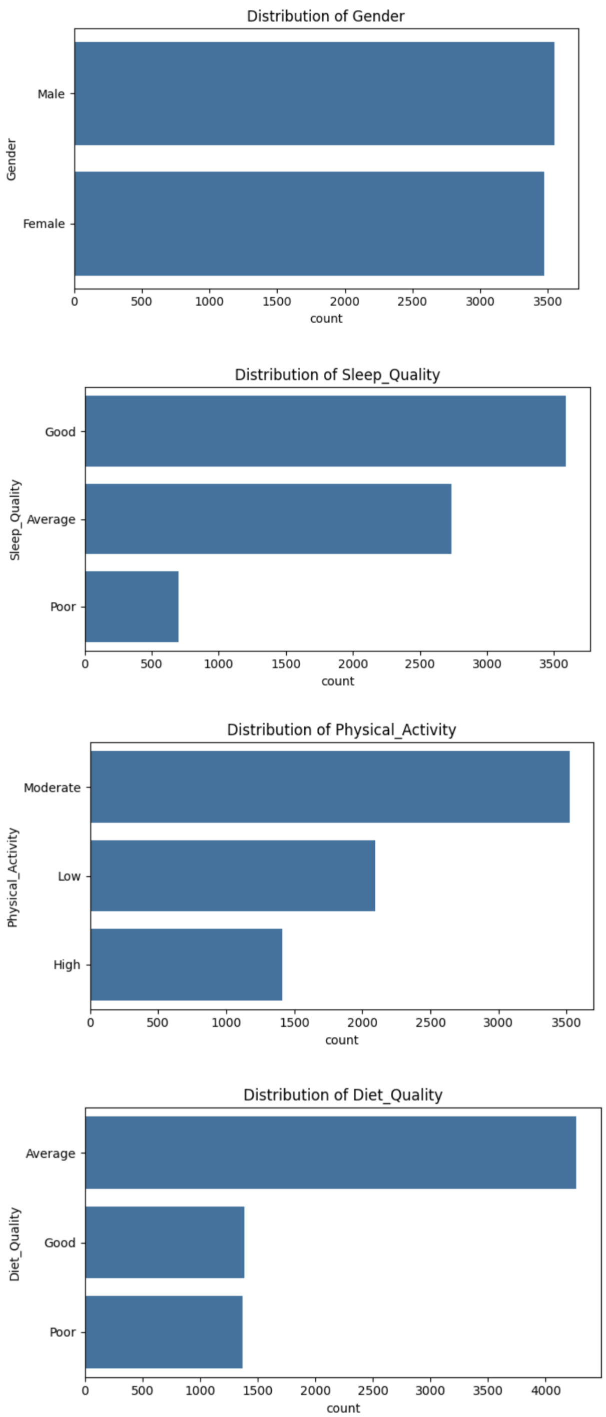

Figure 1: Distribution of key categorical features in the dataset

3.3 Feature Selection and Preprocessing

After understanding the dataset, we moved to feature selection and preprocessing. The dataset contains both numerical and categorical features, so some preparation is needed before using it in machine learning models.

For feature selection, it is important to note that stress, anxiety, and depression scores were used to construct the target variable. Including these same variables as input features can lead to data leakage, artificially inflating model performance.

Therefore, the predictive modeling primarily focused on behavioral, academic, and lifestyle-related features, while the mental health scores were used only for target construction.

For preprocessing, categorical values were converted into numerical form using encoding, because machine learning models cannot directly work with text values. Numerical features were scaled so that all values are in a similar range. This is important because some models are sensitive to feature scale. We also split the dataset into training and testing sets. The training data is used to train the model, and the testing data is used to evaluate performance on unseen data. Another important step was handling missing or inconsistent values, if any, so that the dataset is clean and suitable for modeling. Overall, these steps help in improving the quality of the data and make sure the models perform better.

3.4 Handling Class Imbalance

One of the main issues we found during EDA was class imbalance. There are fewer high-risk students compared to low-risk students, which can affect model performance. To handle this, we used SMOTE, which creates synthetic samples for the minority class. This helps balance the dataset and improves the model's ability to detect high-risk students. SMOTE was only applied on the training data to avoid data leakage.

3.5 Modeling Approach

In this project, we used multiple models like Logistic Regression, Random Forest, Support Vector Machine (SVM), and Gradient Boosting. First, we trained baseline models on the original dataset to see how they perform. Then, after applying SMOTE, we trained the models again to see if the results improve. The goal is to compare different models and see which one performs better for this problem.

3.6 Evaluation Strategy

To evaluate the models, we used metrics like accuracy, precision, recall, and F1-score. Accuracy alone is not enough here because of class imbalance. A model can get high accuracy but still fail to detect high-risk students. So we also focus on recall and F1-score, since they give a better idea of how well the model is performing. Using multiple metrics helps us understand the model performance better.

4 Technical Details

4.1 Data Preprocessing

The dataset consists of both numerical and categorical variables. To prepare the data for machine learning models, several preprocessing steps were applied.

First, categorical features such as gender, relationship status, and residence type were converted into numerical representations using label encoding or one-hot encoding, depending on the feature type. This transformation is necessary because most machine learning models cannot directly process categorical data. Next, numerical features such as sleep quality, physical activity, and academic scores were scaled using standardization. This ensures that all features are on a similar scale, which is particularly important for models such as Logistic Regression and Support Vector Machines. The dataset was then split into training and testing sets using a standard train-test split strategy. The training data was used for model learning, while the testing data was reserved for evaluating model performance on unseen data.

4.2 Target Construction and Leakage Consideration

The target variable was constructed by averaging stress, anxiety, and depression scores and applying a threshold to create a binary classification label (high-risk vs. low-risk). It is important to note that these mental health scores were used to define the label and therefore must be carefully handled during feature selection. Including these same variables as predictors may lead to data leakage, artificially inflating model performance. To ensure a realistic prediction setting, the primary focus was placed on behavioral, academic, and lifestyle features as predictors.

4.3 Handling Class Imbalance

As observed in the earlier analysis, the dataset has a clear class imbalance, where low-risk students are more than high-risk students (approximately 71% vs 29%). This imbalance can cause models to perform poorly on the minority class, especially in detecting high-risk students. To address this issue, we used SMOTE (Synthetic Minority Over-sampling Technique). SMOTE works by generating synthetic samples for the minority class instead of simply duplicating existing data points. This helps create a more balanced dataset and allows the model to learn better decision boundaries for the minority class.

In our implementation, SMOTE was applied as part of a pipeline using `ImbPipeline`. This ensures that SMOTE is applied only on the training data within each fold during cross-validation, which helps prevent data leakage. We also avoided using class weights together with SMOTE. Using both at the same time can lead to overcompensation (double balancing), which may distort the learning process.

After applying SMOTE, all models were retrained and evaluated again. The results showed that the models were able to detect high-risk students better, especially in terms of recall and F1-score. For example, Logistic Regression with SMOTE achieved the best performance with a test F1-score of 0.423 and recall of 0.608, which is significantly better compared to the baseline models. Overall, handling class imbalance using SMOTE played an important role in improving the model's ability to identify high-risk students.

4.4 Models

Multiple classification models were used to evaluate performance:

- Logistic Regression: A linear baseline model for binary classification.
- Random Forest: An ensemble method using multiple decision trees.
- Support Vector Machine (SVM): A model that finds optimal decision boundaries.
- Gradient Boosting: An ensemble method that builds models sequentially.
- XGBoost: An optimized gradient boosting implementation.

These models were chosen to represent a range of linear and non-linear learning approaches.

4.5 Model Selection and Threshold Tuning

Instead of relying on a single model, a dynamic selection approach was used. The best-performing model was selected based on the highest F1-score on the test set. Additionally, threshold tuning was applied to adjust the decision boundary of the classifier. Since identifying high-risk students is more important than overall accuracy, the threshold was adjusted to improve recall and F1-score rather than accuracy alone.

4.6 Evaluation Metrics

Due to class imbalance, accuracy alone is not a reliable metric. Therefore, the following evaluation metrics were used:

- Precision: Measures how many predicted high-risk students are actually high-risk.
- Recall: Measures how many actual high-risk students are correctly identified.
- F1-score: Harmonic mean of precision and recall.
- ROC-AUC: Measures the model's ability to distinguish between classes.

Among these, F1-score was considered the primary metric, as it balances precision and recall.

5 Experiments

5.1 Data Description

The dataset consists of 7,022 student records with 20 features, including demographic, academic, and lifestyle-related attributes. A key observation from the dataset is class imbalance. The number of low-risk students is significantly higher than high-risk students, which affects model performance and requires special handling techniques.

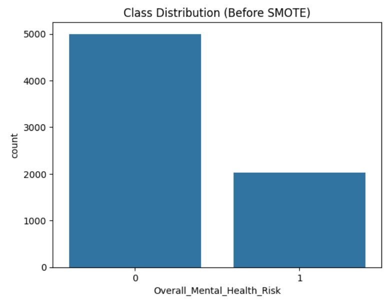

Figure 2: Distribution of low-risk and high-risk students.

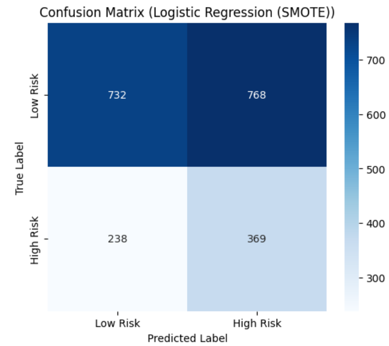

Figure 3: Confusion matrix of the best-performing model.

5.2 Evaluation Metrics

The models were evaluated using accuracy, precision, recall, and F1-score. Due to class imbalance, F1-score was chosen as the primary metric.

5.3 Baseline Models

Baseline models were trained on the original dataset without applying SMOTE. These models generally achieved high accuracy but showed poor recall for the high-risk class.

5.4 Results Before SMOTE

Before applying SMOTE, the models were biased toward predicting the majority class. This resulted in:

- High accuracy due to dominance of the low-risk class
- Low recall for high-risk students
- Poor F1-scores

5.5 Results After SMOTE

After applying SMOTE, the models showed improved ability to detect high-risk students. The key improvements observed were:

- Increased recall for high-risk class
- Improved F1-scores
- More balanced classification behavior

Table 1: Model performance comparison after applying SMOTE

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.523	0.325	0.608	0.423
Random Forest	0.689	0.342	0.087	0.139
SVM	0.607	0.299	0.270	0.284
Gradient Boosting	0.701	0.293	0.028	0.051
XGBoost	0.66	0.329	0.173	0.227

5.6 Analysis

Although SMOTE improved recall and F1-score, overall model performance remained moderate. The best-performing model achieved an F1-score of approximately 0.457.

This suggests that predicting mental health risk using indirect survey features is inherently challenging. Possible reasons include:

- Weak correlation between input features and target variable
- Noise and subjectivity in survey responses
- Limited feature representation of mental health conditions

These results highlight the importance of feature engineering and data quality in predictive modeling tasks.

5.7 Case Study

A closer examination of misclassified instances revealed that some students with moderate lifestyle factors were incorrectly classified. This indicates that the model struggles to capture subtle patterns in the data, further supporting the need for richer features and more advanced modeling approaches.

6 Related Work

The application of machine learning techniques in mental health prediction has gained significant attention in recent years. Several studies have explored the use of behavioral, demographic, and psychological features to identify individuals at risk of mental health conditions such as anxiety and depression.

Previous work has shown that supervised learning models, including Logistic Regression, Support Vector Machines (SVM), Random Forests, and Gradient Boosting, can be effectively used for mental health classification tasks. For instance, Pedregosa et al. (2011) highlighted the effectiveness of such models within the Scikit-learn framework for handling structured datasets. These models are widely used due to their interpretability and ability to capture both linear and non-linear relationships.

One of the most common challenges in mental health datasets is class imbalance. Chawla et al. (2002) introduced the Synthetic Minority Over-sampling Technique (SMOTE), which has become a standard approach for improving classification performance on imbalanced datasets. SMOTE generates synthetic samples for the minority class, helping models better learn patterns associated with rare but important cases, such as high-risk individuals.

In the context of student mental health, several studies have demonstrated that features such as sleep quality, academic performance, and social support play a role in predicting mental health outcomes. However, many existing approaches rely heavily on direct psychological indicators (e.g., stress, anxiety, and depression scores) as input features.

A key limitation identified in the literature is that such direct indicators are not always available in real-world deployment scenarios. In contrast, this project focuses on predicting mental health risk using indirect features such as lifestyle, academic, and demographic attributes, while using psychological scores only for target construction. This makes the problem more realistic but also more challenging.

Overall, prior work establishes the effectiveness of machine learning for mental health prediction and highlights the importance of handling class imbalance. However, it also suggests that the choice of features plays a critical role in model performance, motivating the approach taken in this project.

7 Conclusion

This project focused on predicting students' mental health risk using machine learning techniques. One of the main challenges we faced was the class imbalance in the dataset, where high-risk students were significantly fewer than low-risk students. This led baseline models to achieve high accuracy while failing to properly identify high-risk cases. To resolve this issue, SMOTE was applied to balance the dataset, along with proper evaluation metrics such as F1-score and recall. The results showed a marked improvement in detecting high-risk students after applying these techniques. Overall, this project shows that accuracy alone is not enough for evaluating models in imbalanced classification problems. Metrics such as F1-score provide a more reliable evaluation. Using SMOTE and proper evaluation strategies significantly improved the model's effectiveness, making it more suitable for real-world mental health risk prediction

References

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Code Availability

The implementation of this project can be accessed through the following Google Colab link: https://colab.research.google.com/drive/1rJuIHQPHZEGQUrfA4pXPjD9yE92qJABk?usp=sharing